

TCE Institutional Activity Intelligence and Evaluation System

Final Case-Study Report (Thiagarajar College of Engineering, Madurai)

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1. Executive Summary

Presents the purpose, method, principal results, and data-quality verdict of the Activity Intelligence System in five lines.

****Measured results (this build):****

Metric	Value
Total activities loaded	8
Verified activities	0
Open review tasks	0
Classification macro-F1 (rule-based, held-out)	0.7347
Department detection accuracy	0.8
Entity extraction F1	0.8

2. Introduction and Objectives

Problems addressed (scattered institutional activity records), the objectives of collection-extraction-classification-analytics across a free toolchain.

3. Case Study Context: Thiagarajar College of Engineering

Institutional context: 14 departments, multi-stakeholder outreach, public events pages as the primary lawful source.

****Reference data:**** 14 departments; 24 categories; 8 stakeholder groups; 5 academic years (2020-2021 ... 2024-2025).

4. Related Work and Literature Review

Positioning relative to institutional analytics dashboards, keyword baselines vs. TF-IDF multi-label classification, rule-based NLQ.

5. Methodology Overview

14-phase build: schema, collection, cleaning, extraction, classification, review, cross-reference, analytics, API, UI, NLQ, evaluation, reporting.

6. System Architecture

SQLite relational core, Python/Flask backend, React/Vite frontend, rule-based NLP with spaCy + scikit-learn.

7. Data Sources and Collection

Verified public TCE pages (robots.txt-respecting crawler); events page as the primary lawful feed.

****Collection outcome:****

Item	Value
Raw archive JSON snapshots	4
Raw archive records	26
Unique raw events	10

8. Data Cleaning and Validation

Unicode/whitespace/punctuation normalisation, HTML stripping, date standardisation, validation flags.

9. Deduplication Strategy

FuzzyWRatio title similarity, date consistency; auto-merge only at ≥ 95 with matching dates, else pending review.

****Dedup outcome (latest pipeline run):****

Item	Value
Duplicate pairs auto-merged (≥ 95 + same date)	0
Duplicate pairs pending review (≥ 70)	0

10. Information Extraction

regex retreat in dates/venue/organizer, spaCy NER for persons/orgs, dictionary+fuzzy department matching.

11. NLP Multi-label Classification

24 institutional categories; rule-based keyword baseline vs TF-IDF logistic/SVM/NB multi-label classifiers.

****Model comparison (held-out test split, Phase 5 / 13):****

Model	macro-F1	micro-F1
Rule-Based Keyword Baseline	0.7347 / 0.8312	
TF-IDF + LogisticRegression (OvR)	0.2917 / 0.4091	
TF-IDF + LinearSVC (OvR)	0.25 / 0.3721	
TF-IDF + MultinomialNB (OvR)	0.0 / 0.0	
Selected model	Rule-Based Keyword Baseline	
Training corpus size	110	

12. Model Selection and Evaluation

Held-out macro-F1 comparison; the rule-based baseline selected with documented, reproducible numbers.

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13. Department Detection

Single/multiple/institution-wide/unknown classification from name + alias matching with institution-wide markers.

14. Stakeholder Detection

Keyword-driven association of activities to the 8 stakeholder groups.

15. Human-in-the-Loop Review

Low-confidence queue, field edit, merge, approve/reject, full review_history audit trail.

16. LinkedIn Cross-Reference

Free, legitimate workflow: search-term building, manual verification subset, honest 'Not Checked' defaults.

17. Analytics and KPIs

Overview, year, department, category, stakeholder, LinkedIn, and data-quality aggregates, all computed live.

18. Flask REST API Design

Pagination, validation, parameterised SQL, structured logging, consistent JSON errors.

****Exposed endpoints:**** `/api/activities``, `/api/activities/:id``, `/api/years``, `/api/departments``, `/api/categories``, `/api/stakeholders``, `/api/analytics/{overview, yearly, departments, categories, stakeholders, linkedin, data-quality}``,

`/api/search`, `/api/review`, `POST /api/review/:id/{edit,approve,reject,linkedin}`, `POST /api/collection/run`, `POST /api/query`.

19. React Dashboard

Vite + React + recharts; global filters; KPI cards and charts driven entirely by /api/analytics/* responses.

20. Natural-Language Query System

Plug-in safe, parameterised templates from intent-filters; no freeform SQL; 'unmapped year' answered honestly as 0.

21. Five-Year Trend Analysis

Seeded academic years with honest handling of events beyond the mapped window.

Yearly breakdown (mapped academic years):

Academic Year	Activities	Verified
2020-2021	0 / 0	
2021-2022	0 / 0	
2022-2023	0 / 0	
2023-2024	0 / 0	
2024-2025	0 / 0	

22. Department-wise Analysis

Involvement and primary counts per department, with multi-department reconciliation.

Department involvement (top rows):

Department	Involvement	Primary
Automobile Engineering	0 / 0	
Chemistry	0 / 0	
Chemical Engineering	0 / 0	
Civil Engineering	0 / 0	
Computer Science and Engineering	0 / 0	
Electronics and Communication Engineering	0 / 0	
Electrical and Electronics Engineering	0 / 0	
English and Humanities	0 / 0	
Information Technology	0 / 0	
Applied Mathematics and Computational Sciences	0 / 0	

23. Category-wise Analysis

Classification counts, primary counts, mean confidence per category.

Categories by classification count (top rows):

Category	Count	Primary	Avg conf
Workshop	8 / 0 / None		
Achievement and Award	0 / 0 / None		
Alumni Event	0 / 0 / None		
Campus Life	0 / 0 / None		
Clubs and Chapters	0 / 0 / None		
Conference	0 / 0 / None		
Cultural Event	0 / 0 / None		
Faculty Development Programme	0 / 0 / None		

Guest Lecture	0 / 0 / None		
Hackathon	0 / 0 / None		

24. Stakeholder-wise Analysis

Involvement counts, primary counts, mean confidence per stakeholder group.

****Stakeholders:****

Stakeholder	Involvement	Primary	Avg conf
Alumni	0 / 0 / None		
Community and Society	0 / 0 / None		
Faculty	0 / 0 / None		
Government and Agencies	0 / 0 / None		
Industry	0 / 0 / None		
Parents	0 / 0 / None		
Non-Teaching Staff	0 / 0 / None		
Students	0 / 0 / None		

25. LinkedIn Visibility Analysis

Match-status breakdown; coverage measured only over the manually verified subset.

****LinkedIn match-status breakdown:****

Status	Count
Manual-verification coverage ratio	0

26. Data Quality Assessment

Field completeness, unknown category/department counts, low-confidence records, review backlog, collection failures.

****Completeness (fraction of fields filled):****

Field	Filled / total
date_present	3 / 8
venue_present	0 / 8
organizer_present	0 / 8
description_present	8 / 8
source_present	8 / 8
confidence_present	0 / 8
no_mapped_academic_year	8 / 8

****Flags:****

Item	Value
Collection errors open	0
Review backlog	0
LinkedIn unmatched/unchecked	0
Records with no mapped academic year	8

27. Evaluation and Validation

Reproducible NLP metrics (macro/micro F1, department/stakeholder accuracy, entity P/R/F) plus pipeline reconciliation.

Metric	Value
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Classification macro-F1	0.7347
Classification micro-F1	0.8312
Department accuracy	0.8
Stakeholder accuracy	0.8
Entity precision	0.6667
Entity recall	1.0
Entity F1	0.8

28. Limitations

Small labeled corpus, single-year 2026 feed for the case study, LinkedIn manual-only workflow, no LLM dependency.

29. Ethical and Data-Integrity Considerations

No fabricated records; only verified public sources; no LinkedIn scraping; audit history on every mutation.

30. Future Work

Collecting more years, growing real labeled data, LLM-only parsing step for NLQ, Postgres migration on free tier, RBAC for reviewers.

31. Conclusion

The system meets the case-study objective with a free, transparent, reproducible toolchain.

****Verdict:**** a free, transparent, reproducible pipeline that loads 8 real activities, classifies them with macro-F1 0.7347, and stands fully auditable via its review_history audit trail.